Lesson 13 Teacher's Key: A Hit That Reacts

Scratch - Instructor Reference Only

VIDEO GAME CREATION

What This Lesson Teaches

Sequencing a visual reaction BEFORE finalizing game-state logic using Scratch's "wait" block, and spawning a fully independent visual effect via broadcast - the first lesson where the enemy's removal is deliberately delayed relative to when it was hit.

Level 1 - Full Worked Solution

- The Explosion sprite is a completely separate, self-contained script - it never touches score, lives, or wave variables, and its own "wait 0.2 seconds -> hide" lifecycle needs no help from the Enemy.
- "broadcast [explosion]" fires and moves on immediately - Scratch broadcasts never block the sender, which is exactly why the explosion can be triggered without disturbing the flash-then-wait sequence that follows it.
- Moving the win-check/delete block below "wait 0.1 seconds" is the only structural change to existing logic - the logic itself is byte-for-byte the same as Lesson 12, just running slightly later.

Why This Structure

- Keeping the explosion as a broadcast-triggered, fully independent script (rather than something the Enemy waits on) means a slow or buggy explosion animation could never accidentally freeze the actual game logic - a good habit even at this small scale.
- This is the first lesson where 'wait' pauses something that MATTERS (delaying a state change) rather than just pacing a visual (Lesson 2's beep-beep) - worth naming that distinction explicitly.

Level 2 / Level 3 Talking Points

- Ask students to trace, step by step, what happens between a bullet touching an enemy and the enemy actually disappearing now - flash, wait, THEN the same Lesson 6-12 logic runs. A good comprehension check for sequencing.
- For strong students: ask what would happen if "broadcast [explosion]" were accidentally placed AFTER "wait 0.1 seconds" instead of before it - the explosion would appear 0.1 seconds late, slightly out of sync with the flash. Not a bug exactly, but worth noticing as a timing choice.

Common Student Mistakes

- Placing the win-check/delete block BEFORE the wait instead of after, which defeats the whole point (the enemy would still vanish instantly).
- Forgetting to reset "set [color] effect to 0" somewhere in the enemy's reset script, causing a revived/respawned enemy to start already tinted.
- Accidentally duplicating the score/win-check lines when moving them below the wait, causing double-counting.